Spring Testing Exercises

Exercise 1: Basic Unit Test for Service

CalculatorService.java

```
import org.springframework.stereotype.Service;

@Service
public class CalculatorService {

    public int add(int a, int b) {
        return a + b;
    }
}
```

CalculatorServiceTest.java

```
import static org.junit.jupiter.api.Assertions.assertEquals;

import org.junit.jupiter.api.Test;

public class CalculatorServiceTest {

    CalculatorService service = new CalculatorService();

    @Test
    void testAdd() {
        assertEquals(10, service.add(4, 6));
    }
}
```

Exercise 2: Mock Repository in Service Test

UserServiceTest.java

```
import static org.junit.jupiter.api.Assertions.*;
import static org.mockito.Mockito.*;

import java.util.Optional;

import org.junit.jupiter.api.Test;
import org.mockito.InjectMocks;
```

```
import org.mockito.Mock;
import org.mockito.MockitoAnnotations;

public class UserServiceTest {

    @Mock
    private UserRepository userRepository;

    @InjectMocks
    private UserService userService;

    public UserServiceTest() {
        MockitoAnnotations.openMocks(this);
    }

    @Test
    void testGetUserById() {

        User user = new User();
        user.setId(1L);
        user.setName("John");

        when(userRepository.findById(1L))
                .thenReturn(Optional.of(user));

        User result = userService.getUserById(1L);

        assertEquals("John", result.getName());
    }
}
```

Exercise 3: REST Controller Testing

UserControllerTest.java

```
import static org.mockito.Mockito.*;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
```

```
import org.springframework.boot.test.autoconfigure.web.servlet.WebMvcTest;
import org.springframework.boot.test.mock.mockito.MockBean;
import org.springframework.test.web.servlet.MockMvc;

@WebMvcTest(UserController.class)
public class UserControllerTest {

   @Autowired
   MockMvc mockMvc;

   @MockBean
   UserService userService;

   @Test
   void testGetUser() throws Exception {

      User user = new User();
      user.setId(1L);
      user.setName("John");

      when(userService.getUserById(1L))
            .thenReturn(user);

      mockMvc.perform(get("/users/1"))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.name")
            .value("John"));
   }
}
```

Exercise 4: Integration Test

```
import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;

@SpringBootTest
public class IntegrationTest {
```

```
    @Autowired
    private UserService userService;

    @Test
    void testIntegration() {

        User user = new User();
        user.setName("John");

        User saved = userService.saveUser(user);

        assertNotNull(saved);
    }
}
```

Exercise 5: Test POST Endpoint

```
import static org.mockito.Mockito.*;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

import org.junit.jupiter.api.Test;

public class UserPostControllerTest {

    @Autowired
    MockMvc mockMvc;

    @MockBean
    UserService userService;

    @Test
    void testCreateUser() throws Exception {

        User user = new User();
        user.setId(1L);
        user.setName("John");

        when(userService.saveUser(any(User.class)))
```

```
            .thenReturn(user);

        mockMvc.perform(post("/users")
            .contentType("application/json")
            .content("""
                {
                  "name":"John"
                }
                """))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.name")
            .value("John"));
    }
}
```

Exercise 6: Service Exception Handling

```
import static org.junit.jupiter.api.Assertions.*;
import static org.mockito.Mockito.*;

import java.util.Optional;
import java.util.NoSuchElementException;

import org.junit.jupiter.api.Test;

public class UserExceptionTest {

    @Test
    void testUserNotFound() {

        UserRepository repo = mock(UserRepository.class);

        when(repo.findById(1L))
            .thenReturn(Optional.empty());

        assertThrows(
            NoSuchElementException.class,
            () -> {
                throw new NoSuchElementException();
            });
```

```
    }
}
```

Exercise 7: Custom Repository Query

```
import static org.junit.jupiter.api.Assertions.*;

import java.util.List;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;

@DataJpaTest
public class RepositoryTest {

    @Autowired
    UserRepository repository;

    @Test
    void testFindByName() {

        User user = new User();
        user.setName("John");

        repository.save(user);

        List<User> users =
              repository.findByName("John");

        assertEquals(1, users.size());
    }
}
```

Exercise 8: Controller Exception Handler

```
import static org.mockito.Mockito.*;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.*;

import java.util.NoSuchElementException;
```

```
import org.junit.jupiter.api.Test;

public class ExceptionHandlerTest {

   @Autowired
   MockMvc mockMvc;

   @MockBean
   UserService userService;

   @Test
   void testExceptionHandler() throws Exception {

      when(userService.getUserById(1L))
            .thenThrow(new NoSuchElementException());

      mockMvc.perform(get("/users/1"))
            .andExpect(status().isNotFound())
            .andExpect(content()
            .string("User not found"));
   }
}
```

Exercise 9: Parameterized Test

```
import static org.junit.jupiter.api.Assertions.*;

import org.junit.jupiter.params.ParameterizedTest;
import org.junit.jupiter.params.provider.ValueSource;

public class ParameterizedCalculatorTest {

   CalculatorService service =
         new CalculatorService();

   @ParameterizedTest
   @ValueSource(ints = {2,4,6,8,10})
   void testEvenNumbers(int number) {
```

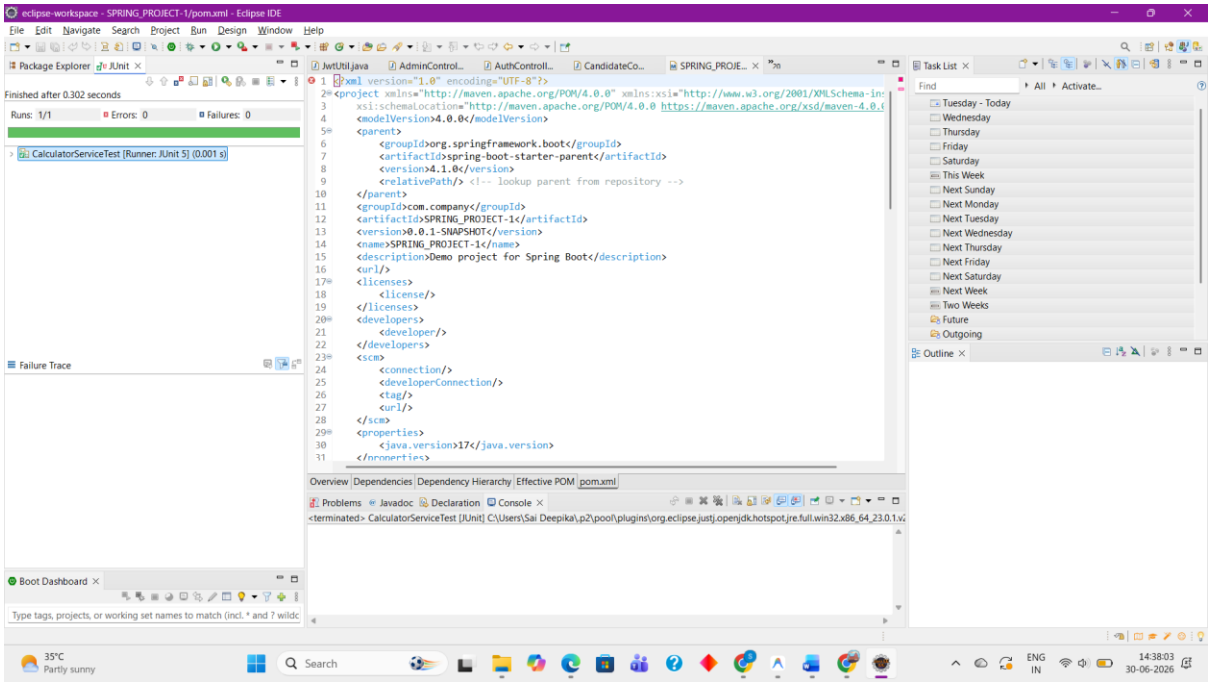
```
        assertEquals(0,
                number % 2);
    }
}
```

Required Maven Dependencies

Add these to your pom.xml:

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-test</artifactId>
    <scope>test</scope>
</dependency>

<dependency>
    <groupId>com.h2database</groupId>
    <artifactId>h2</artifactId>
    <scope>test</scope>
</dependency>
```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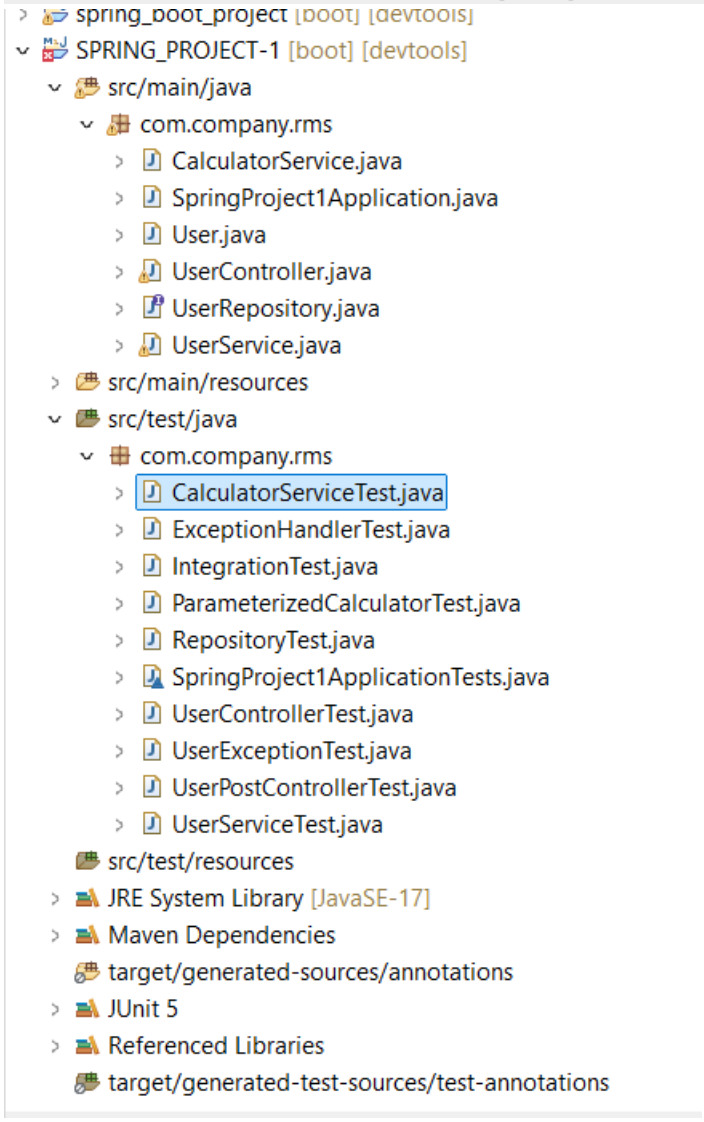
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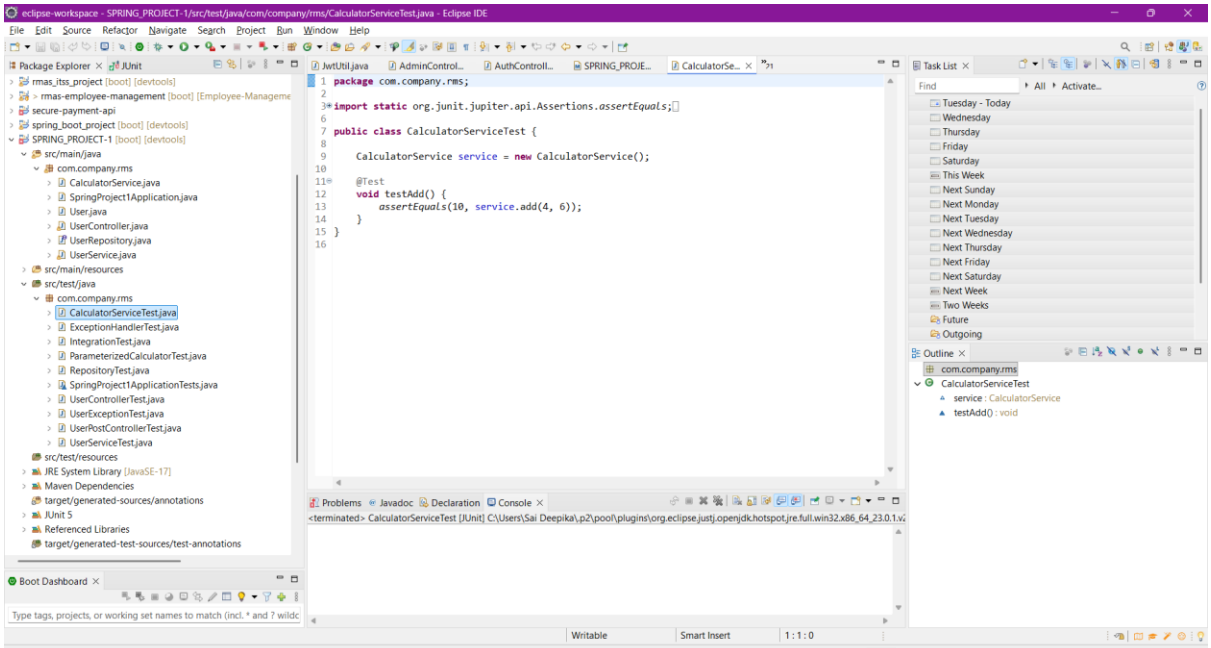
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1 package com.company.rms;
3 import static org.junit.jupiter.api.Assertions.assertEquals;
7 public class CalculatorServiceTest {
9 CalculatorService service = new CalculatorService();
11 @Test
12 void testAdd() {
13 assertEquals(10, service.add(4, 6));
14 }
15 }
Problems Javadoc Declaration Console
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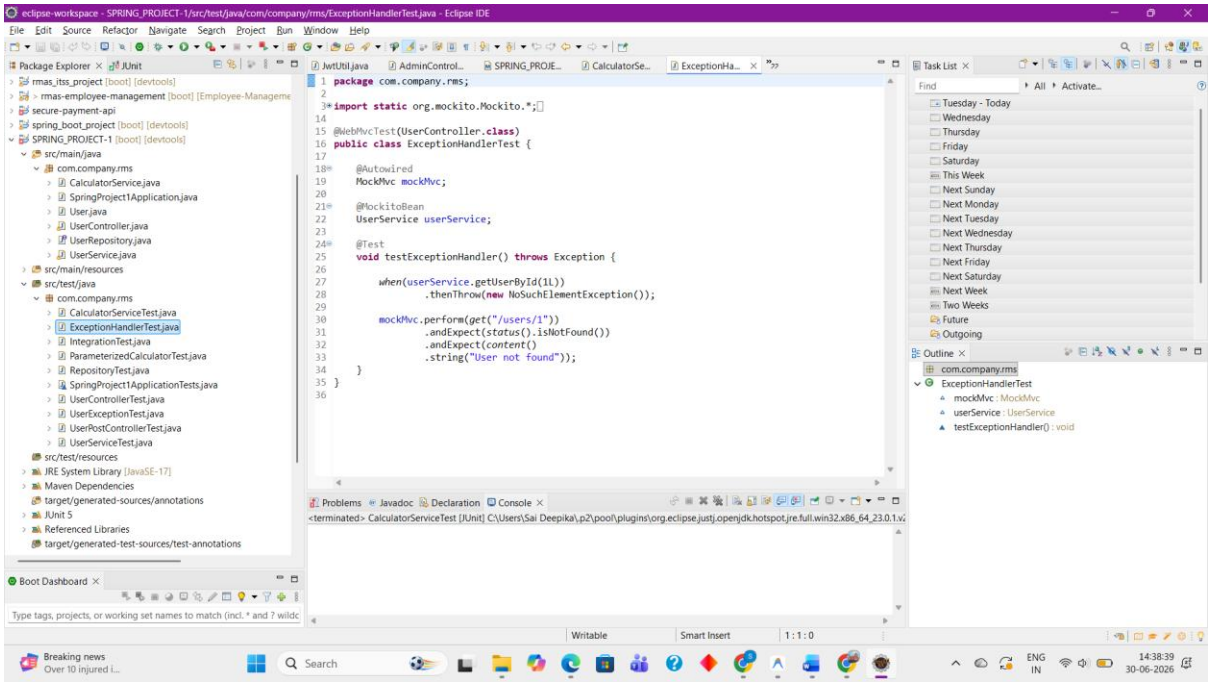
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3 import static org.mockito.Mockito.*;
15 @WebMvcTest(UserController.class)
16 public class ExceptionHandlerTest {
18 @Autowired
19 MockMvc mockMvc;
21 @MockitoBean
22 UserService userService;
24 @Test
25 void testExceptionHandler() throws Exception {
27 when(userService.getUserById(1L))
28 .thenThrow(new NoSuchElementException());
30 mockMvc.perform(get("/users/1"))
31 .andExpect(status().isNotFound())
32 .andExpect(content()
33 .string("User not found"));
34 }
35 }
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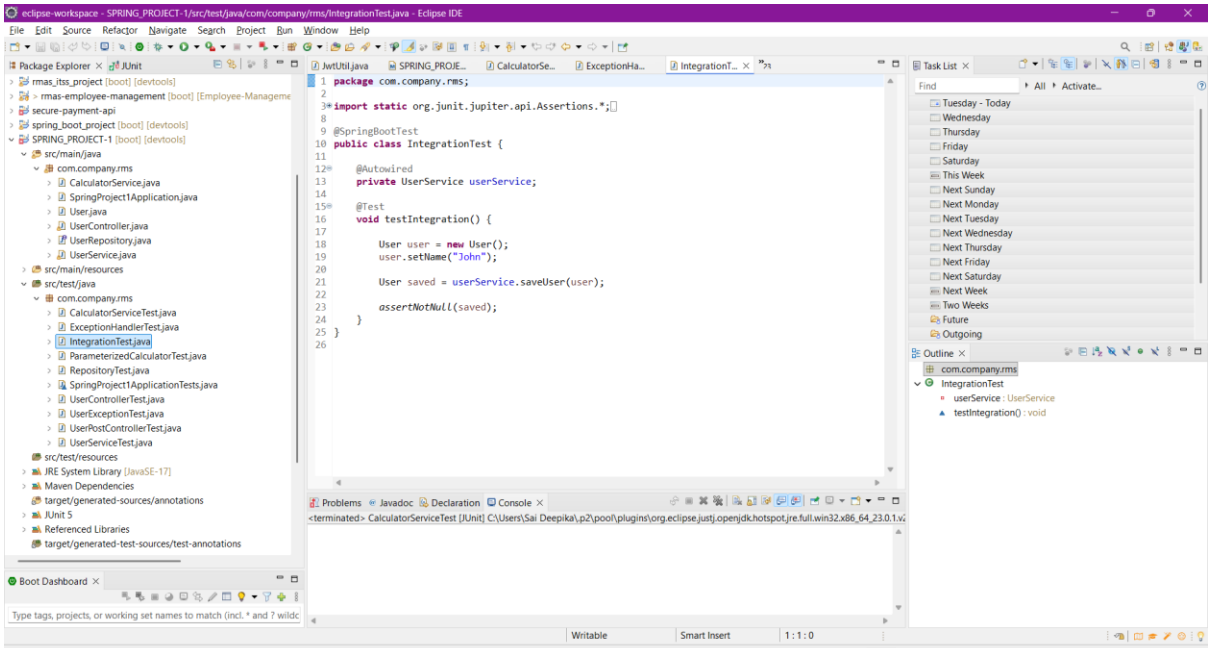
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1 package com.company.rms;
2
3 import static org.junit.jupiter.api.Assertions.*;
8
9 @SpringBootTest
10 public class IntegrationTest {
11
12 @Autowired
13 private UserService userService;
14
15 @Test
16 void testIntegration() {
17
18 User user = new User();
19 user.setName("John");
20
21 User saved = userService.saveUser(user);
22
23 assertNotNull(saved);
24 }
25 }
26
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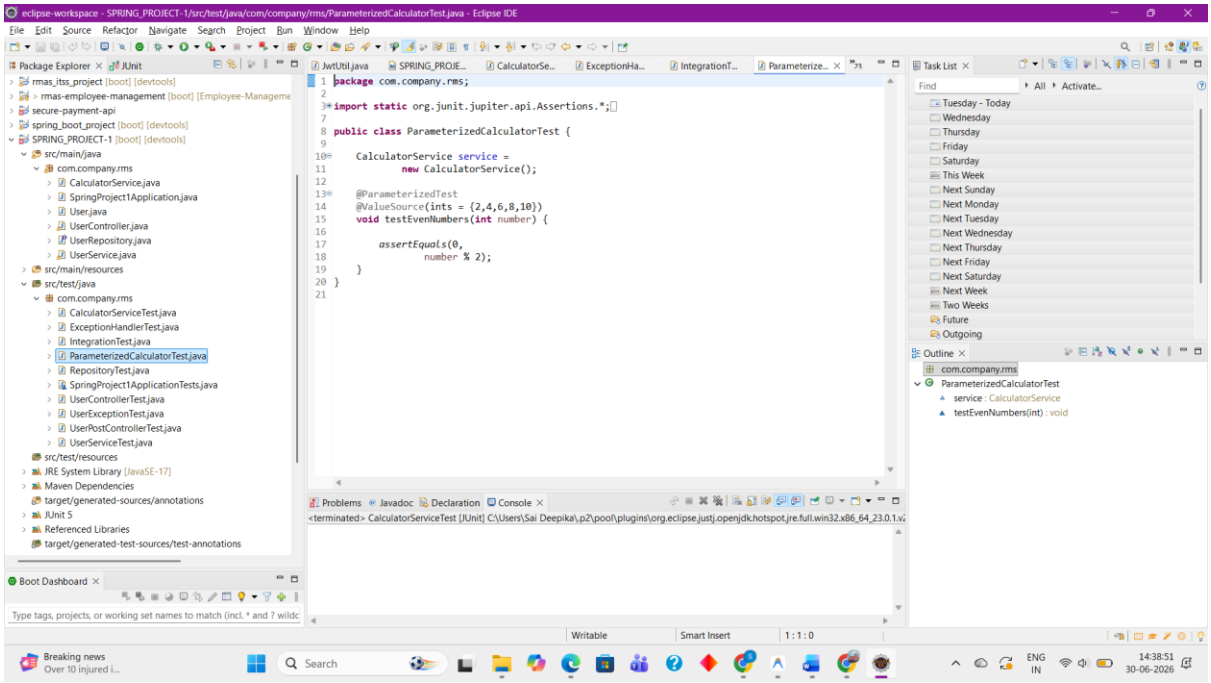
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11 new CalculatorService();
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14 @ValueSource(ints = {2,4,6,8,10})
15 void testEvenNumbers(int number) {
16
17 assertEquals(0,
18 number % 2);
19 }
20 }
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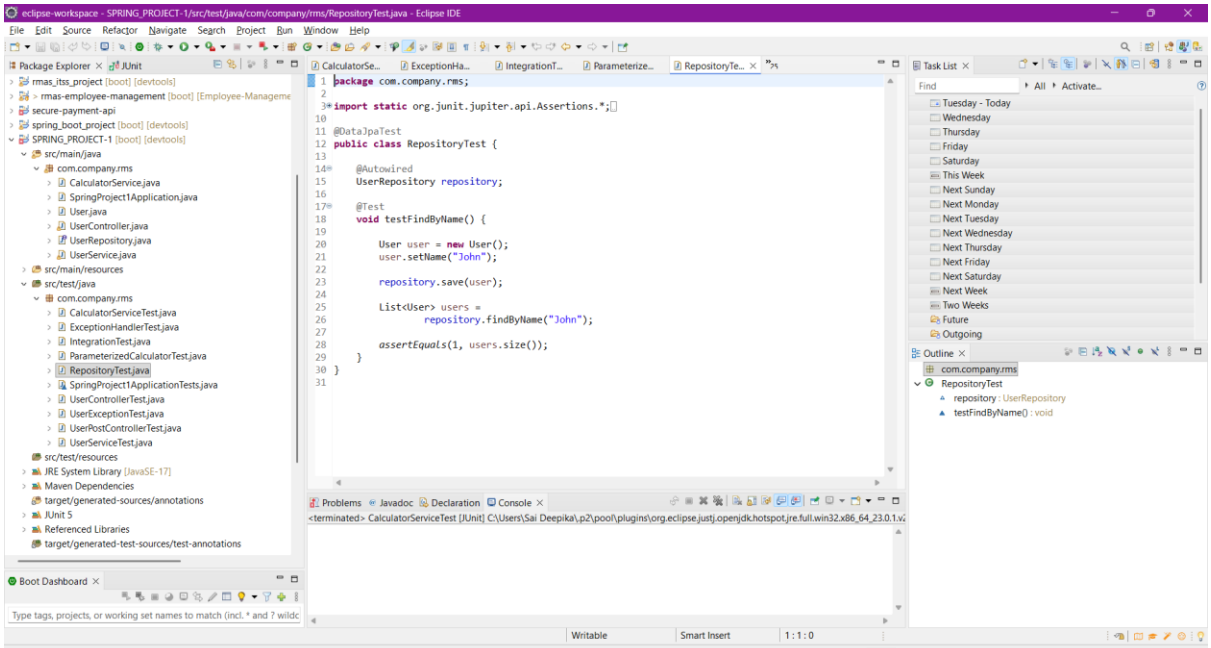
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14 @Autowired
15 UserRepository repository;
16
17 @Test
18 void testFindByName() {
19
20 User user = new User();
21 user.setName("John");
22
23 repository.save(user);
24
25 List<User> users =
26 repository.findByName("John");
27
28 assertEquals(1, users.size());
29 }
30 }
31
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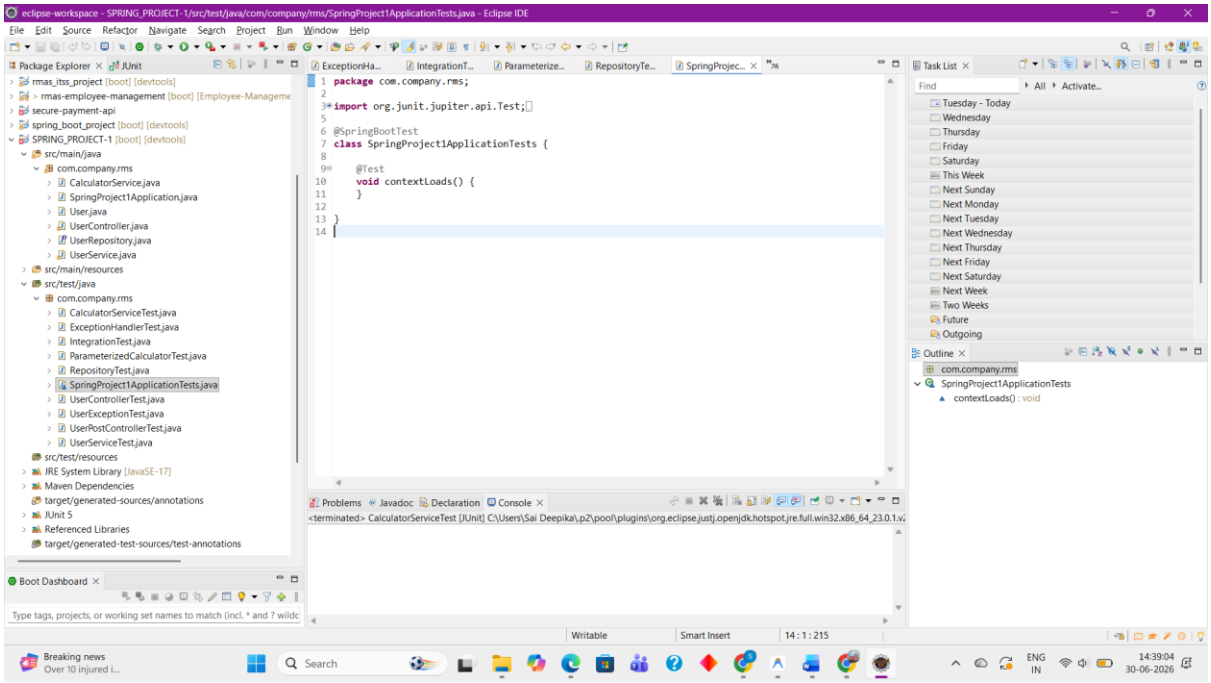
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1 package com.company.rms;
2
3 import org.junit.jupiter.api.Test;
5
6 @SpringBootTest
7 class SpringProject1ApplicationTests {
8
9 @Test
10 void contextLoads() {
11 }
12
13 }
14
Problems
Javadoc
Declaration
Console
<terminated> CalculatorServiceTest [JUnit] C:\Users\Sai Deepika\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v
Task List
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com.company.rms
SpringProject1ApplicationTests
contextLoads() : void
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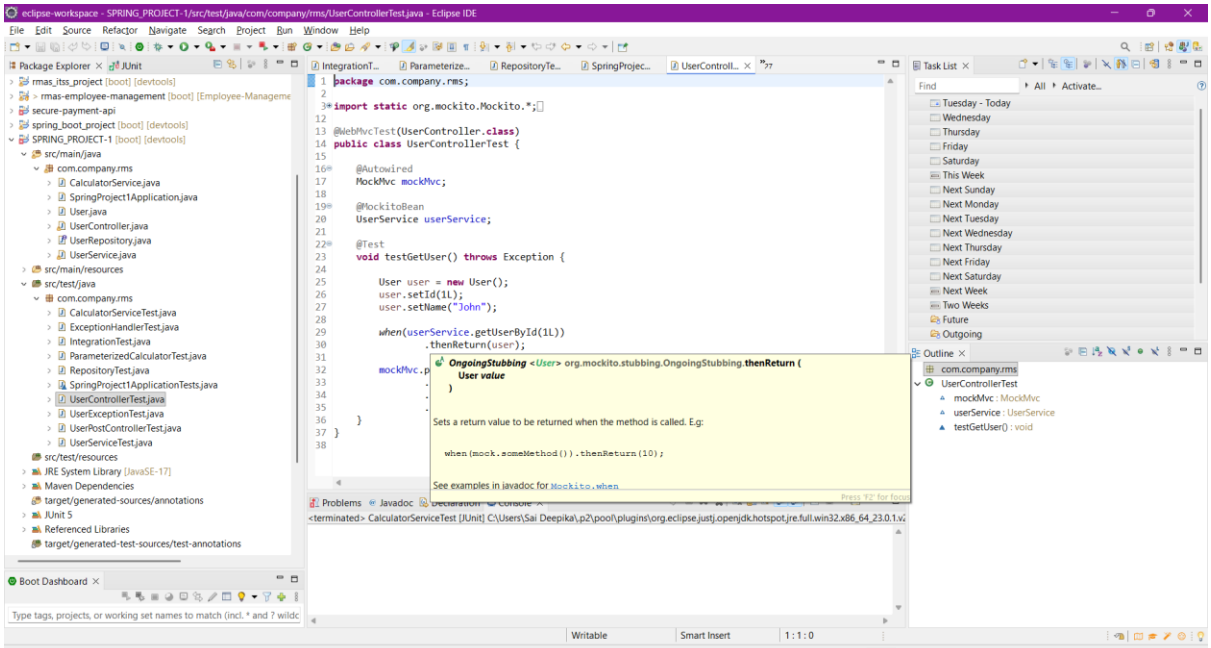
eclipse-workspace - SPRING_PROJECT-1/src/test/java/com/company/rms/UserControllerTest.java - Eclipse IDE
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JUnit
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UserRepository.java
UserService.java
src/main/resources
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target/generated-test-sources/test-annotations
Boot Dashboard
IntegrationT...
Parameterize...
RepositoryTe...
SpringProjec...
UserControll...
1 package com.company.rms;
3 import static org.mockito.Mockito.*;
13 @WebMvcTest(UserController.class)
14 public class UserControllerTest {
16 @Autowired
17 MockMvc mockMvc;
19 @MockitoBean
20 UserService userService;
22 @Test
23 void testGetUser() throws Exception {
25 User user = new User();
26 user.setId(1L);
27 user.setName("John");
29 when(userService.getUserById(1L))
30 .thenReturn(user);
32 mockMvc.p
36 }
37 }
OngoingStubbing <User> org.mockito.stubbing.OngoingStubbing.thenReturn (
User value
)
Sets a return value to be returned when the method is called. E.g:
when(mock.someMethod()).thenReturn(10);
See examples in javadoc for Mockito.when
Press 'F2' for focus
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<terminated> CalculatorServiceTest [JUnit] C:\Users\Sai Deepika\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v
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com.company.rms
UserControllerTest
mockMvc : MockMvc
userService : UserService
testGetUser() : void
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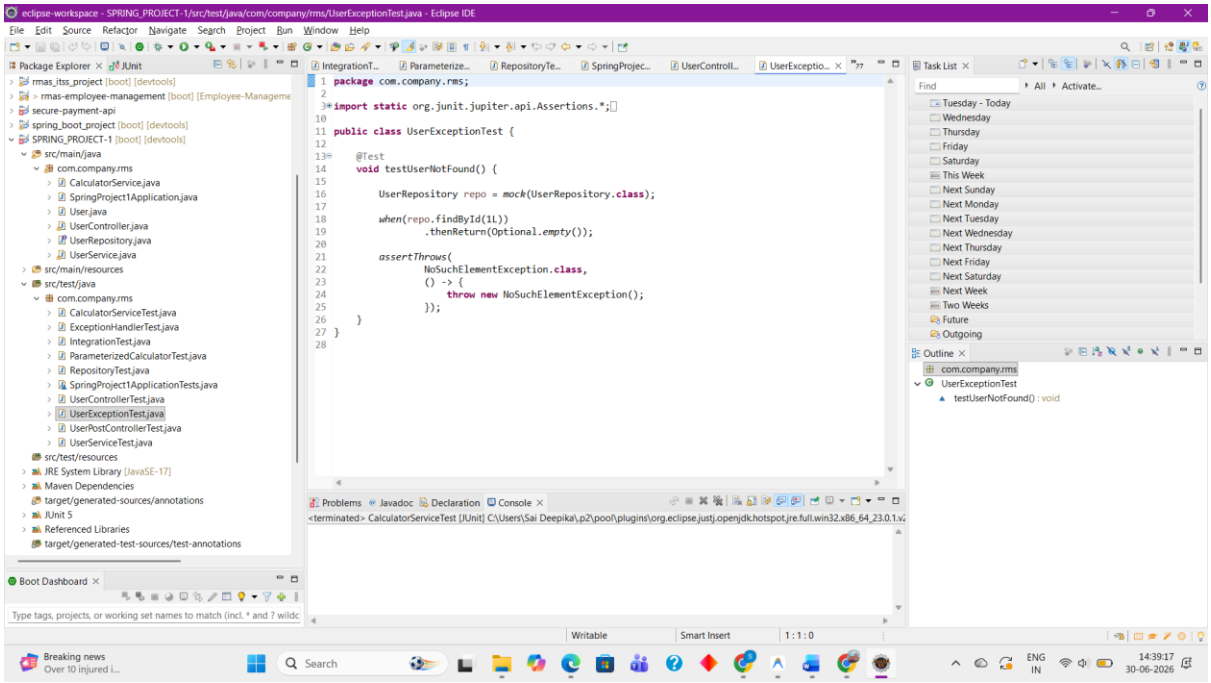
eclipse-workspace - SPRING_PROJECT-1/src/test/java/com/company/rms/UserExceptionTest.java - Eclipse IDE
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UserControll...
UserExceptio...
1 package com.company.rms;
3 import static org.junit.jupiter.api.Assertions.*;
11 public class UserExceptionTest {
13 @Test
14 void testUserNotFound() {
16 UserRepository repo = mock(UserRepository.class);
18 when(repo.findById(1L))
19 .thenReturn(Optional.empty());
21 assertThrows(
22 NoSuchElementException.class,
23 () -> {
24 throw new NoSuchElementException();
25 });
26 }
27 }
Problems
Javadoc
Declaration
Console
<terminated> CalculatorServiceTest [JUnit] C:\Users\Sai Deepika\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v
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com.company.rms
UserExceptionTest
testUserNotFound() : void
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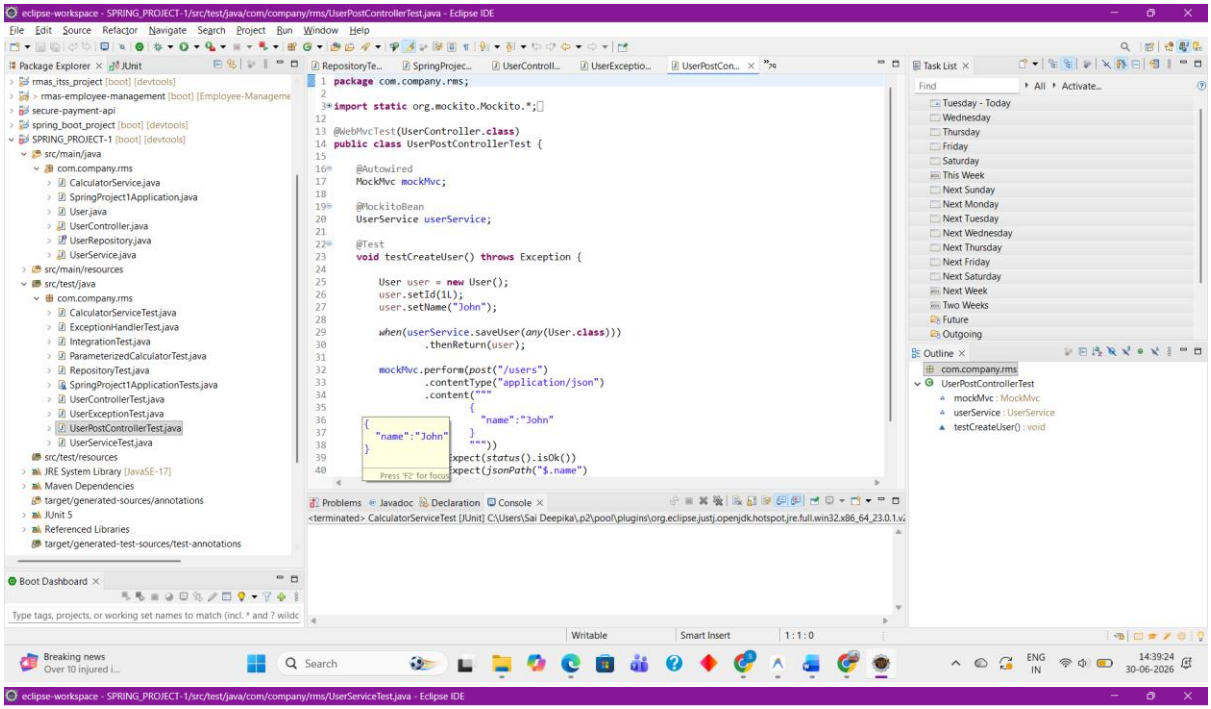
eclipse-workspace - SPRING_PROJECT-1/src/test/java/com/company/rms/UserPostControllerTest.java - Eclipse IDE
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RepositoryTe...
SpringProjec...
UserControll...
UserExceptio...
UserPostCon...
1 package com.company.rms;
3 import static org.mockito.Mockito.*;
13 @WebMvcTest(UserController.class)
14 public class UserPostControllerTest {
16 @Autowired
17 MockMvc mockMvc;
19 @MockitoBean
20 UserService userService;
22 @Test
23 void testCreateUser() throws Exception {
25 User user = new User();
26 user.setId(1L);
27 user.setName("John");
29 when(userService.saveUser(any(User.class)))
30 .thenReturn(user);
32 mockMvc.perform(post("/users")
33 .contentType("application/json")
34 .content("""
35 {
36 "name":"John"
37 }
38 """))
39 .andExpect(status().isOk())
40 .andExpect(jsonPath("$.name")
{
"name":"John"
}
Press 'F2' for focus
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testCreateUser() : void
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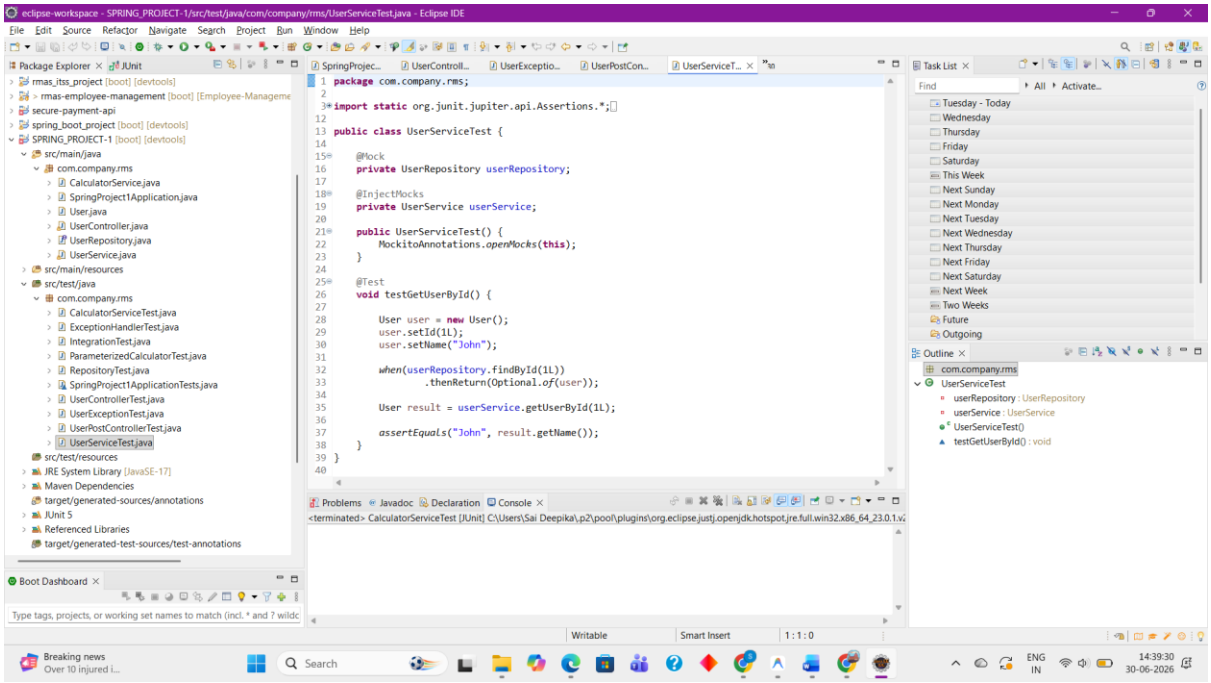
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Boot Dashboard
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SpringProjec...
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UserExceptio...
UserPostCon...
UserServiceT...
1 package com.company.rms;
3 import static org.junit.jupiter.api.Assertions.*;
13 public class UserServiceTest {
15 @Mock
16 private UserRepository userRepository;
18 @InjectMocks
19 private UserService userService;
21 public UserServiceTest() {
22 MockitoAnnotations.openMocks(this);
23 }
25 @Test
26 void testGetUserById() {
28 User user = new User();
29 user.setId(1L);
30 user.setName("John");
32 when(userRepository.findById(1L))
33 .thenReturn(Optional.of(user));
35 User result = userService.getUserById(1L);
37 assertEquals("John", result.getName());
38 }
39 }
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<terminated> CalculatorServiceTest [JUnit] C:\Users\Sai Deepika\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_23.0.1.v
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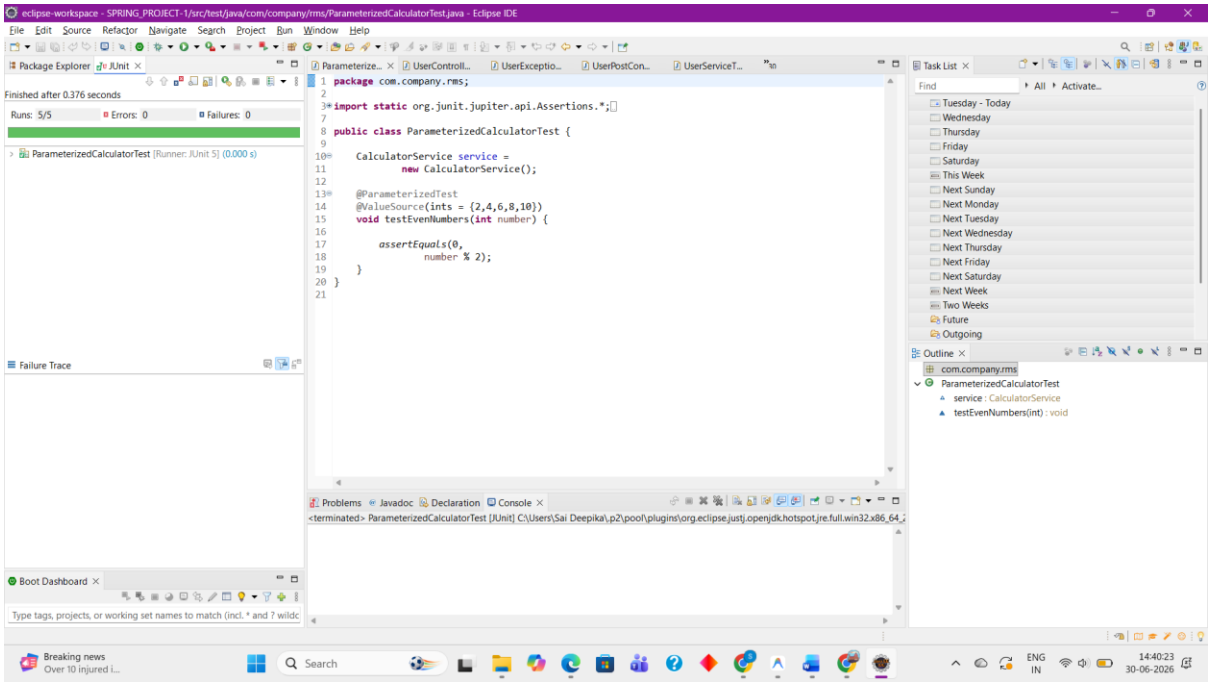
eclipse-workspace - SPRING_PROJECT-1/src/test/java/com/company/rms/ParameterizedCalculatorTest.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer JUnit
Finished after 0.376 seconds
Runs: 5/5 Errors: 0 Failures: 0
ParameterizedCalculatorTest [Runner: JUnit 5] (0.000 s)
Failure Trace
Boot Dashboard
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Parameterize... UserControll... UserExceptio... UserPostCon... UserServiceT...
1 package com.company.rms;
2
3 import static org.junit.jupiter.api.Assertions.*;
7
8 public class ParameterizedCalculatorTest {
9
10 CalculatorService service =
11 new CalculatorService();
12
13 @ParameterizedTest
14 @ValueSource(ints = {2,4,6,8,10})
15 void testEvenNumbers(int number) {
16
17 assertEquals(0,
18 number % 2);
19 }
20 }
21
Problems Javadoc Declaration Console
<terminated> ParameterizedCalculatorTest [JUnit] C:\Users\Sai Deepika\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_
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ParameterizedCalculatorTest
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testEvenNumbers(int) : void
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